MINI PROJECT

Kth Largest Element in a Binary Search Tree (BST)

1. Problem Statement

Develop a C++ program to find the Kth largest element in a Binary Search Tree (BST). The program accepts a collection of distinct integer values, inserts them into a BST, takes the position K from the user, and finds the element at that position when the values are arranged in descending order. It must also check whether the entered value of K is valid.

2. Introduction

A Binary Search Tree is a non-linear, hierarchical data structure used to store values in an organised form. It consists of nodes, where each node contains data and links to its left and right children. In a standard BST, values smaller than a node are inserted into its left subtree, while values greater than the node are inserted into its right subtree. The same rule applies recursively to every subtree.

The Kth largest element is the value that appears at position K when all values are listed from largest to smallest. For example, the largest element is at position 1, the second largest is at position 2, and so on. This operation is useful in ranking, result processing, employee salary analysis, and other applications where a particular rank must be identified.

This project uses reverse inorder traversal, which visits the right subtree first, then the root node, and finally the left subtree (Right → Root → Left). Since larger BST values are located toward the right, this traversal visits the values in descending order. A counter is updated at every visited node. When the counter reaches K, the current node contains the required Kth largest value. This avoids sorting the full collection separately.

Objectives

• Construct a BST from input values.
• Understand recursive traversal.
• Find a ranked element and validate K.
• Analyse the time and space complexity of the solution.